CAL POLY

CANTILEVER BEAM LAB

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LAB OBJECTIVES

- CONDUCT A SINE SWEEP, RANDOM NOISE, AND FREE VIBRATION TEST ON A CANTILEVER BEAM. COMPARE THE ADVANTAGES AND DISADVANTAGES OF USING EACH FOR FINDING NATURAL FREQUENCIES.
- EXPERIMENTALLY PRODUCE BODE PLOTS AND FREQUENCY RESPONSE GRAPHS FROM THE SPECTRUM ANALYZER AND OSCILLOSCOPE
  - UTILIZE ACCELEROMETERS TO GATHER RECORD DATA
- DEDUCE THE CANTILEVER BEAM'S FUNDAMENTAL NATURAL FREQUENCY AND CALCULATE THE DAMPING COEFFICIENT
  - APPLY THE LOG DECREMENT EQUATION
- RECORD AND ANALYZE FOOTAGE OF THE BEAM'S VIBRATION TO UNDERSTAND ITS PHYSICAL BEHAVIOR

EXPERIMENTAL HARDWARE

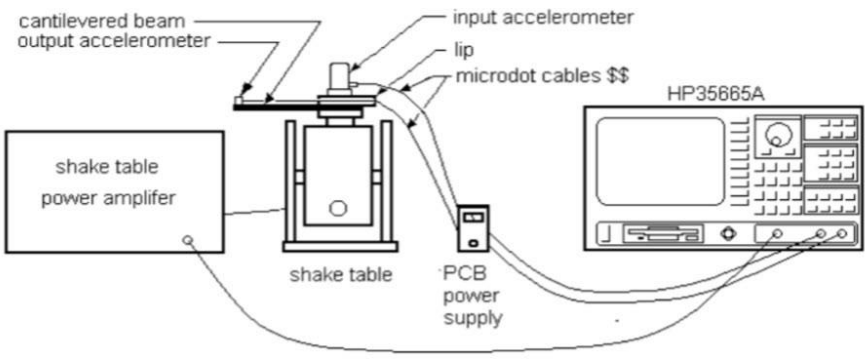
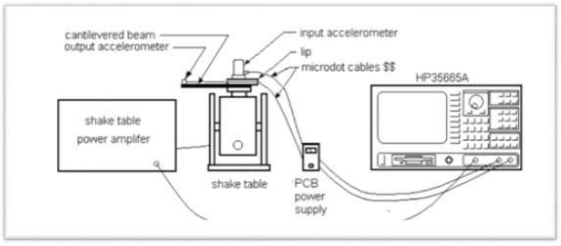


FIGURE: OVERALL EXPERIMENT SETUP AND HARDWARE
(KEYSIGHT OSCILLOSCOPE NOT SHOWN)

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SINE SWEEP TEST (EXPERIMENTAL SETUP)

- CANTILEVER BEAM MOUNTED ON A SHAKE TABLE.
- ACCELEROMETERS AT THE BASE AND TIP TO MEASURE VIBRATIONS.
- SPECTRUM ANALYZER USED TO GENERATE A SINE WAVE INPUT AND CAPTURE THE SYSTEM'S RESPONSE.
- WIRING INCLUDES MICRODOT AND BNC CABLES TO CONNECT SENSORS TO THE SPECTRUM ANALYZER.
- REFER TO FIGURE

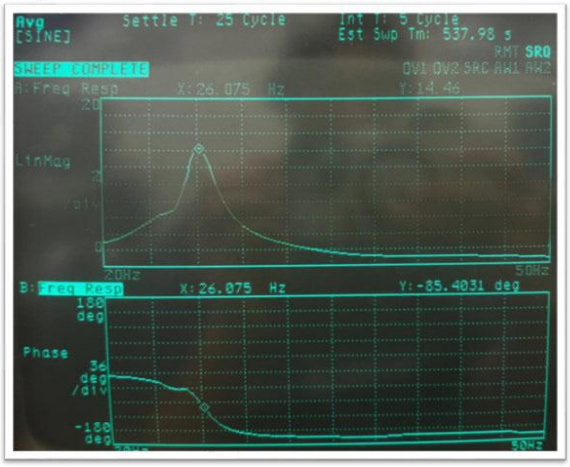


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SINE SWEEP TEST (FRF GRAPHS)



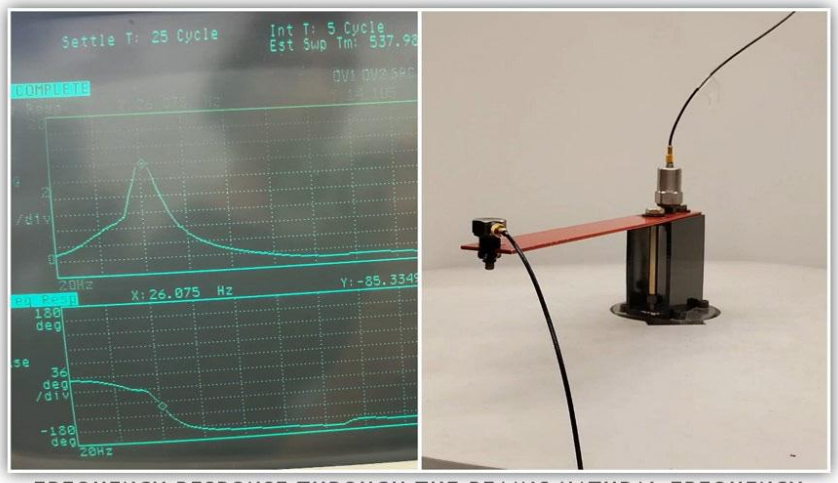
TEST 1 (10-310 Hz) RESULTS:
Wn1[Hz]: 26.5 Hz
Wn2[Hz]: 214 Hz



TEST 2 (20-50 Hz) RESULTS:
Wn1[Hz]: 26.075 Hz

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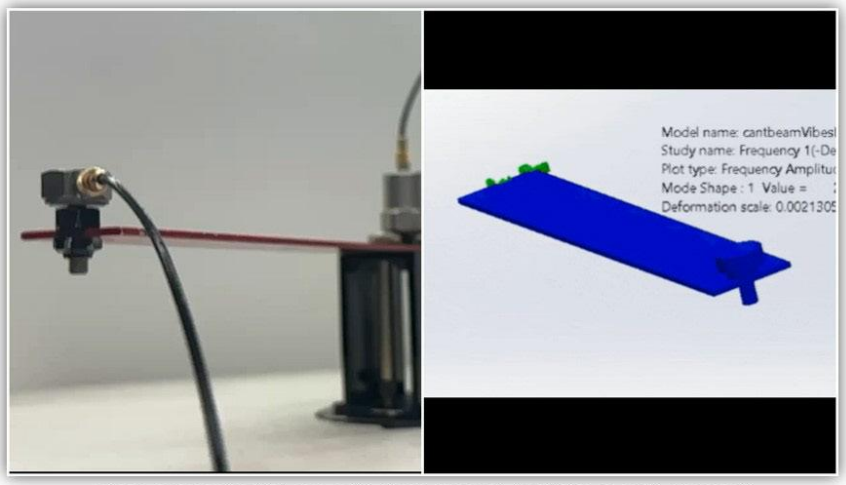
SINE SWEEP TEST (FUNDAMENTAL NATURAL FREQUENCY)



FREQUENCY RESPONSE THROUGH THE BEAM'S NATURAL FREQUENCY

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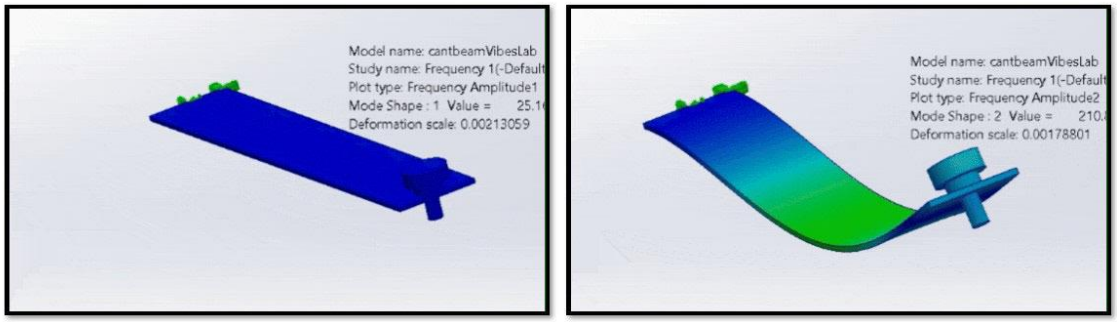
RESPONSE TO FORCED VIBRATION AT RESONANCE



FIRST RESONANT FREQUENCY VIBRATION IN SLOW MOTION

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MODE SHAPES OF CANTILEVER BEAM

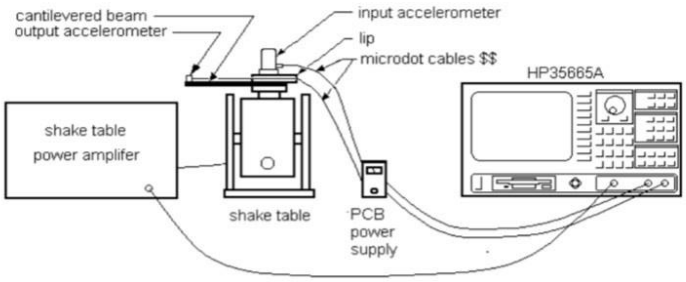


MODE 1 (LEFT) AND MODE 2 (RIGHT) VIBRATIONS IN SOLIDWORKS

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RANDOM NOISE TEST (EXPERIMENTAL SETUP)

THE SPECTRUM ANALYZER SENDS A RANDOM NOISE SIGNAL COVERING A WIDE RANGE OF FREQUENCIES TO THE SHAKE TABLE. THE CANTILEVER BEAM IS VIBRATED WITH RANDOM NOISE, WHICH EXCITES ALL POSSIBLE NATURAL FREQUENCIES SIMULTANEOUSLY. THE ACCELEROMETERS AT THE BASE AND TIP RECORD THE VIBRATIONAL RESPONSE. THIS INPUT IS DISPLAYED ON THE SPECTRUM ANALYZER AS A FREQUENCY RESPONSE FUNCTION (FRF) OR BODE PLOT. THE ONLY DIFFERENCE IN SET UP FROM THE SINE SWEEP TESTS IS THE SETUP OF THE SPECTRUM ANALYZER.

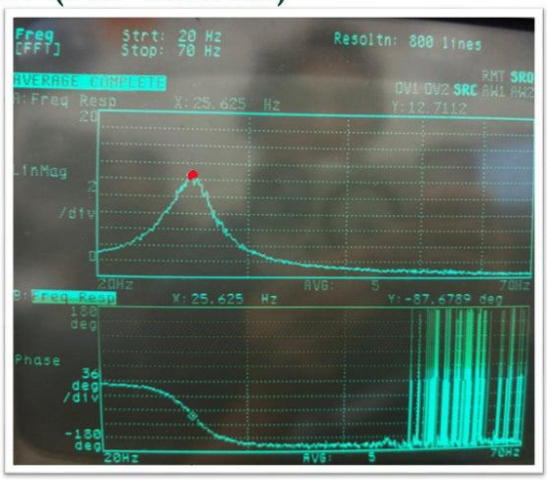


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RANDOM NOISE TEST (FRF GRAPHS)



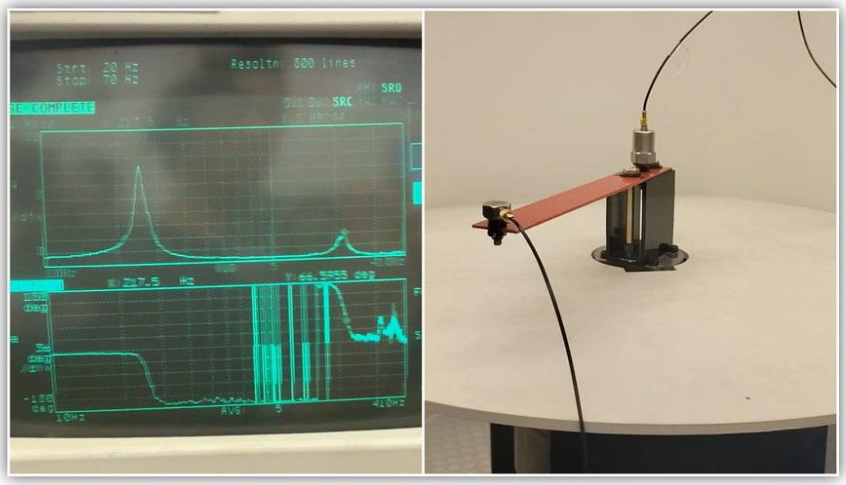
TEST 1 (10-310 Hz) RESULTS:
Wn1[Hz]: 25 Hz
Wn2[Hz]: 217.5 Hz



TEST 2 (20-50 Hz) RESULTS:
Wn1[Hz]: 25.625 Hz

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RANDOM NOISE TEST (FUNDAMENTAL NATURAL FREQUENCY)

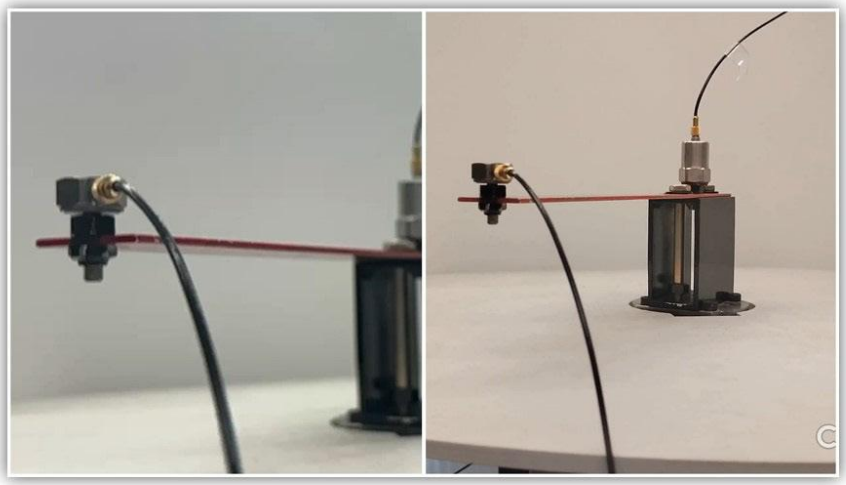


SLOW MOTION VIBRATION COMPARISON

FIRST RESONANT FREQUENCY (LEFT) VS RANDOM NOISE (RIGHT)

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SLOW MOTION RESPONSE COMPARISON



SLOW MOTION VIBRATION COMPARISON

FIRST RESONANT FREQUENCY (LEFT) VS RANDOM NOISE (RIGHT)

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TRANSIENT RESPONSE TEST AND LOG DECREMENT METHOD

THE LOGARITHMIC DECREMENT METHOD IS USED TO CALCULATE THE **DAMPING RATIO (Z)** FROM THE OBSERVED DECAY IN AMPLITUDE: **Δ=LN(X1/(XN+1))D;**

- **X1**=AMPLITUDE OF THE FIRST OSCILLATION PEAK,
- **XN+1**=AMPLITUDE OF THE N+1TH PEAK.

THE DAMPING RATIO IS THEN FOUND USING THE RELATIONSHIP: **Z=Δ/2Π**

THEORETICAL ANALYSIS:

THE **DAMPED NATURAL FREQUENCY (Ω_D)** IS CALCULATED BASED ON THE NATURAL FREQUENCY **(ΩN)** AND THE DAMPING RATIO: **ΩD=ΩN * SQRT(1-Z^2)**

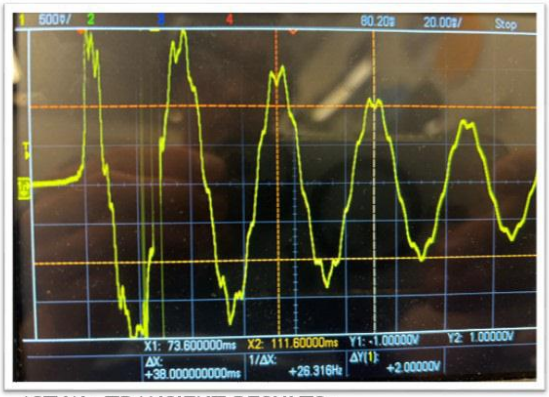
- **ΩN** IS THE NATURAL FREQUENCY FOUND FROM PREVIOUS TESTS (SINE SWEEP OR RANDOM NOISE).
- **ΩD** REPRESENTS THE TRUE FREQUENCY OF OSCILLATION DURING FREE VIBRATION, ACCOUNTING FOR ENERGY LOSS.

SIGNIFICANCE:

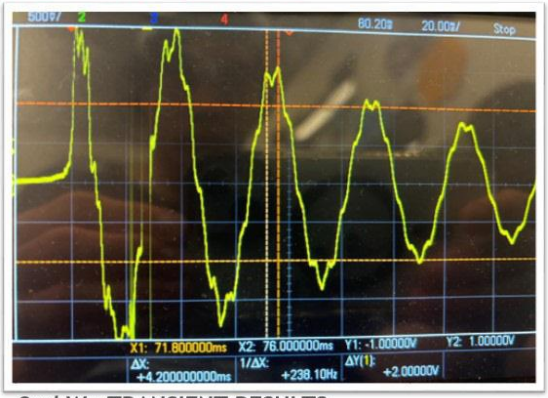
- THE TRANSIENT RESPONSE ILLUSTRATES HOW QUICKLY ENERGY DISSIPATES IN THE SYSTEM.
- A HIGHER DAMPING RATIO **(Z)** MEANS FASTER DECAY OF OSCILLATIONS, INDICATING HIGHER ENERGY DISSIPATION.
- THIS METHOD HELPS DETERMIN THE RESONANCE STABILITY IN BEAMS.

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TRANSIENT RESPONSE TEST RESULTS



1ST Wn TRANSIENT RESULTS:
Wn1[Hz]: 26.316 Hz



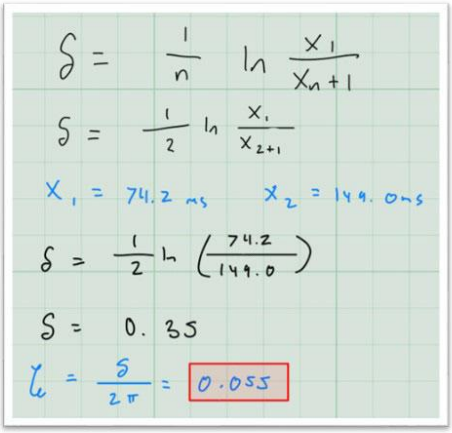
2nd Wn TRANSIENT RESULTS:
Wn2[Hz]: 238.10 Hz

RESULTS COMPARISON TO SWEEP AND RANDOM NOISE TEST :

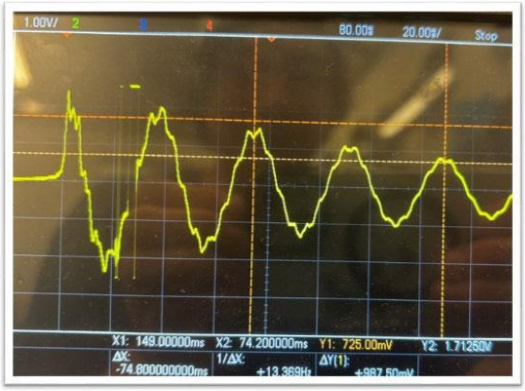
THE RESULTS OF THE TESTS ARE ALL VERY SIMILAR, HOWEVER THERE IS A SLIGHT DIFFERENCE. WHILE THE FIRST NATURAL FREQUENCY WERE ALL PRETTY MUCH THE SAME, FOR THE TRANSIENT TEST OUR 2ND NATURAL FREQUENCY WAS ROUGHLY 20HZ HIGHER. THOUGH SOME VARIANCE BETWEEN THE TESTS THIS ERROR WAS MOST LIKELY DUE TO MISPLACEMENT OF THE CURSORS DUE TO THE SECOND FREQUENCY PRODUCING A MUCH SMALLER WAVE ON THE ANALYZER MAKING IT HARDER TO PLACE ACCURATE CURSORS.

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TRANSIENT RESPONSE TEXT (DECREMENT METHOD)



DECREMENT METHOD CALCULATION FOR ζ



TRANSIENT RESPONSE WITH CURSORS TO FIND X1 AND Xn+1

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ERROR ANALYSIS

TEST	Wn1(Hz)	ERROR(%)	Wn2(Hz)	ERROR(%)
CALCULATED VALUES	25.169	N/A	210.87	N/A
SINE SWEEP 1	26.5	5.29%	214	1.48%
SINE SWEEP 2	26.075	3.60%	N/A	N/A
RANDOM NOISE 1	25	-0.67%	217.5	3.14%
RANDOM NOISE 2	25.625	1.81%	N/A	N/A
TRANSIENT RESPONSE	26.316	4.56%	238.1	12.91%

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CONCLUSION AND DISCUSSION

- THE FIRST NATURAL FREQUENCY WAS CONSISTENTLY IDENTIFIED ACROSS ALL TESTS (~26.5 HZ), WITH MINOR VARIATIONS OBSERVED IN THE RANDOM NOISE AND TRANSIENT RESPONSE TESTS.
- THE SECOND NATURAL FREQUENCY SHOWED SOME DISCREPANCIES, PARTICULARLY IN THE TRANSIENT RESPONSE TEST (238.10 HZ), LIKELY DUE TO CURSOR PLACEMENT ERRORS AND LOWER SIGNAL AMPLITUDE/CLARITY.
- THE RANDOM NOISE TEST EFFICIENTLY CAPTURED ALL RESONANT FREQUENCIES SIMULTANEOUSLY. IT IS A FAST METHOD, BUT THE NOISE ADDS UNCERTAINTY.
- THE SINE SWEEP TEST IDENTIFIED THE FREQUENCIES MORE PRECISELY WITH CLEARER PEAKS BUT REQUIRED MORE TIME TO EXECUTE.
- THE TRANSIENT RESPONSE TEST EFFECTIVELY MEASURED DAMPING CHARACTERISTICS, REVEALING HOW ENERGY DISSIPATES IN REAL-WORLD SCENARIOS.
- DIFFERENCES BETWEEN THE THREE METHODS HIGHLIGHT THE IMPORTANCE OF METHOD SELECTION BASED ON PRECISION AND TIME CONSTRAINTS.
- THE EXPERIMENT UNDERSCORED THE SIGNIFICANCE OF ACCOUNTING FOR RESONANCE IN STRUCTURAL APPLICATIONS TO PREVENT FAILURE AND FATIGUE.

CHECKLIST

EXPERIMENT 6
CANTILEVER BEAM LAB CHECKLIST

Task Description	X
COMPLETE THE CHECKLIST BELOW AND INCLUDE AT THE END OF LAB REPORT	X
Hardware Setup Completed as per Figure 1	X
Complete Tabular Column for Sine Sweep Test 1	X
Complete Tabular Column for Sine Sweep Test 2	X
Obtain videos V1, V2, & V3	X
Complete Tabular Column for Random Noise Test 1	X
Complete Tabular Column for Random Noise Test 2	X
Obtain videos V4, V5 & V6	X
Obtain the three photos of the transient response (lowest frequency, next higher frequency, and delayed response(Log Decrement))	X
Power Off Spectrum Analyzer	X
Turn off Power to the Shaker Amplifier (Push Button on Front Panel)	X
Unplug the Two Power Cords for the Shaker Amplifier	X
Turn off Power to the Signal Conditioner (Toggle Switch)	X